

REAL ESTATE SALES ANALYSIS USING EXCEL, POWER QUERY, AND POWER BI

MINI PROJECT DOCUMENTATION

1. PROJECT OVERVIEW

The Real Estate Sales Analysis project focuses on analyzing property listing data using Microsoft Excel, Power Query, and Microsoft Power BI. The project demonstrates the complete data analytics process, beginning with data cleaning, followed by data transformation, business analysis, and dashboard creation.

The dataset contains information such as property type, city, locality, listing price, property size, listing status, listing date, and agent details. The cleaned and transformed data is analyzed using Power BI to generate meaningful business insights through interactive dashboards.

This project helps users understand property market trends, compare property prices across cities, evaluate listing performance, and monitor important business metrics using KPI cards and visual reports.

2. PROJECT OBJECTIVES

The objectives of this project are:

- To clean and organize raw real estate data using Microsoft Excel.
 - To identify and remove duplicate records and missing values.
 - To standardize and validate the dataset.
 - To transform the dataset using Power Query.
 - To create DAX measures for business calculations.
 - To design an interactive Power BI dashboard.
 - To create KPI cards for important business metrics.
 - To analyze property prices and listing trends.
 - To generate business insights that support decision-making.
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3. PROBLEM STATEMENT

Real estate companies manage thousands of property listings every day. Analyzing this large volume of data manually is time-consuming and prone to errors. There is a need for an automated dashboard that provides quick insights into property prices, listing status, city-wise distribution, and overall business performance.

This project addresses this challenge by using Excel for data cleaning and Power BI for interactive visualization and analysis.

4. SCOPE OF THE PROJECT

The project covers:

- Data cleaning and preparation
- Data transformation
- KPI calculation
- Dashboard development
- Business analysis
- Interactive reporting

The project does not include predictive analytics or machine learning models.

5. DATASET DESCRIPTION

The dataset contains property listing information.

Fields

- PropertyID
- PropertyType
- Address
- City
- Locality
- Bedrooms
- Bathrooms
- SquareFeet
- ListPrice
- ListingStatus
- ListDate
- AgentID

The dataset is used to analyze pricing trends, property distribution, and listing performance.

6. SOFTWARE AND TOOLS USED

- Microsoft Excel
 - Power Query in Power BI
 - Microsoft Power BI Desktop
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7. METHODOLOGY

The project follows the following workflow:

1. Data Collection
 2. Data Cleaning in Excel
 3. Data Transformation using Power Query
 4. Data Loading into Power BI
 5. Data Modeling
 6. DAX Measure Creation
 7. Dashboard Development
 8. Business Analysis
 9. Report Generation
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8. DATA CLEANING IN EXCEL

The following activities were performed:

- Removed duplicate records.
 - Checked for missing values.
 - Corrected formatting issues.
 - Converted dates into proper format.
 - Applied currency formatting to ListPrice.
 - Created additional calculated columns:
 - Price Category
 - Price per Square Foot
 - Year
 - Month
 - Property Size
 - Converted the dataset into an Excel Table.
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9. DATA TRANSFORMATION USING POWER QUERY

The cleaned dataset was imported into Power Query.

The following transformations were applied:

- Changed data types.
- Removed duplicate records.
- Trimmed unnecessary spaces.
- Renamed columns.
- Validated data consistency.
- Loaded the transformed data into Power BI.

10. DAX MEASURES

The following measures were created:

- Total Revenue
- Total Properties
- Average Price
- Maximum Price
- Minimum Price
- Total Cities
- Total Agents

These measures are displayed as KPI cards in the dashboard.

11. DASHBOARD DESIGN

The Power BI dashboard contains:

KPI Cards

- Total Revenue
- Total Properties
- Average Price
- Maximum Price
- Minimum Price
- Total Cities
- Total Agents

Charts

- Properties by City (Column Chart)
- Listing Status (Pie Chart)
- Monthly Revenue Trend (Line Chart)
- Average Price by Property Type (Bar Chart)

Slicers

- City
- Property Type
- Listing Status
- Year

The dashboard provides interactive filtering and visualization for effective analysis.

12. KEY PERFORMANCE INDICATORS (KPIs)

The project includes the following KPIs:

- Total Revenue
- Total Properties
- Average Property Price
- Highest Property Price
- Lowest Property Price
- Number of Cities
- Number of Agents

These KPIs provide a quick summary of the overall performance of the property listings.

13. BUSINESS INSIGHTS

The dashboard enables users to:

- Identify cities with the highest number of property listings.
 - Compare average property prices across different property types.
 - Analyze monthly revenue trends.
 - Understand the distribution of listing statuses.
 - Evaluate the coverage of properties across multiple cities.
 - Monitor the performance of property listings through interactive filters.
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14. CHALLENGES FACED

The following challenges were encountered during the project:

- Cleaning inconsistent data.
- Handling duplicate records.
- Formatting date and currency fields.
- Creating DAX measures.
- Designing an interactive dashboard.

These challenges were resolved using Excel, Power Query, and Power BI features.

15. CONCLUSION

The Real Estate Sales Analysis project successfully demonstrates the use of Microsoft Excel, Power Query, and Power BI to transform raw data into meaningful business insights.

The project highlights the importance of data cleaning, transformation, KPI development, and dashboard visualization in supporting informed decision-making.

The interactive dashboard enables users to monitor property listings, analyze pricing trends, compare city-wise performance, and evaluate key business metrics effectively.

16. FUTURE SCOPE

Future enhancements may include:

- Integration with live real estate databases.
 - Predictive analytics for property price forecasting.
 - Geographic mapping of property locations.
 - AI-based recommendation systems.
 - Automated dashboard refresh using cloud services.
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17. REFERENCES

- Microsoft Excel Documentation
- Microsoft Power Query Documentation
- Microsoft Power BI Documentation
- Real Estate Dataset used for the project